

Lab-4: Applied Statistical Modeling & Interactive

Web Dashboard (Understanding)

- Create a streamlit an app that creates a UI showing data, graphs, etc
- Used the **Medical insurance Cost** topic's dataset:
insurance.csv dataset from Kaggle
- To run the streamlit setup streamlit environment and then run
- Created a requirements.txt file to setup dependencies like pandas, numpy etc.
- Work flow
Dataset → EDA → Hypothesis Testing → Multiple Linear Regression → Model
Diagnostics → Prediction → Streamlit Dashboard
- The target variable for the dataset is **charges (Y)** medical insurance charges

PART 1: Exploratory Data Analysis (EDA):

Part A: Descriptive Metrics & Visual Exploration:

- Understanding the data
Number of rows: 1338 Number of columns: 7
- 7 variable names
- Data types:

Age	Numerical
Sex	Categorical
Bmi	Numerical
Children	Numerical
Smoker	Categorical
Region	Categorical
Charges	Numerical

- Computed the measures of central tendency (mean, median), dispersion (standard deviation, IQR), skewness, and kurtosis across numerical features (age, BMI, children, charges)
- Plotted graphs: Histogram for distribution plots across different variables, Scatter plot for bivariate variables and correlation heatmap to display the correlation between variables. Each graph has filter options to check for every numerical features. Boxplot to check for outliers using the five number summary.

Part B: Hypothesis Test 1 & Hypothesis Test 2:

- **Hypothesis test 1:** Here we need to pick a categorical variable with 2 values and compare it with a numerical variable and check if there are any significant changes.
- For the dataset, selected the group variable smoking status and comparing it with the charges to check any changes between the groups **Non-smoker's** and **Smoker's**

- Formulate the **H0** that indicates that there is **no significant change** between the variables for both the categories in that case their means are equal belongs to **Normally distributed Population**, **H1** indicates that there is a significant difference between them and their **means are not the same**, does not belong to **Normally distributed Population**.
- Performed the **normality check** on the smoker and non-smoker groups
 1. Smokers as 'Yes' and non-smokers as 'No' the continuous variable charges
 2. Computed Shapiro wilk for each group
 3. The **Shapiro wilk** test for both groups shows that they have been computed close to normal distribution, but the **p value turns out to be <0.000001** that is way far value compared to <0.05 to obtain a normal distribution hence it **rejects the normality for both the hypothesis H0 and H1**.
 4. The less p value indicates that the probability associated with that W under the normality assumption for the data is very small.
- **Levene's test** is computed to check the **spread in the groups**, the mean may be same but the spread may differ, hence we calculate the variance. The value obtained measure the evidence against equal variance assumption. We take the already computed assumption from Shapiro wilk test.
- For this dataset the Levene's test shows higher value hence higher variation but the p value is very low hence the **H0 is rejected** in this case.
- Since the normality check proved to be rejected, used the **Mann-Whitney U test that does not require data to be normally distributed, works on ranks**. Check if the distribution between two groups different or not.
- We use ranks here as there can be skewness as result the magnitude can change and hence focusing on relative ordering of the groups.
- **Hypothesis test 2:** Selected the **One-Way ANOVA test** to check the **distribution of medical charges across the four regions**, the categorical variable is Geographic Region and the numerical variable is charges.
- Computes the variation between different groups, within the group and between the groups.
- **H0: Mean charges are equal across all 4 regions; H1: At least one region has a different mean for charges.**
- One – way ANOVA computes the F-stats, mean across the 4 regions and the result of p value **rejects H0**, and tells that at least one region has different mean for charges but does not tell which region.

PART 2: Statistical Modelling & Diagnostic Checks:

- **Model Formulation:** Used to predict medical charges (charges) from several patient characteristics at the same time.
- Regression used to understand the relation between dependent and one or more independent variables. Here the dependent variable Y is charges and the independent variables X are age, sex, BMI, children, smoker, region.
- Simple linear regression we only have one predictor, whereas for **multiple linear regression** we have more than one predictor.
- We use multiple variables because charges are influenced by many factors simultaneously, for each case we may find a relationship.
- **Ordinary Least Squares (OLS)** used to estimate **regression coefficients** that minimizes the sum of squared residuals, makes the error as small as possible.
- **Parameter Interpretation:** the R^2 value called **coefficient of determination**, measures the proportion of the variation in the dependent variable (charges) that is explained by the predictors in your regression model.
- the value obtained shows the variation in medical charges of the dataset 84.09% indicates that the regression model is good, the remaining predictions are also included in the model.
- **Adjusted R^2** is similar to R^2 , but it accounts for the number of predictors in the model and the sample size.
- **Gauss-Markov Diagnostic Checks:** After fitting the regression model, it is important to investigate whether important regression assumptions are reasonable. The project performs three major diagnostic checks:
 1. **Residuals vs Fitted Values:** A positive residual means the model underpredicted the charge. A negative residual means the model overpredicted the charge. The x-axis represents Fitted/Predicted Charges, and y-axis represents residuals, residuals should appear as a relatively random cloud around zero.
 - **Homoscedasticity** means that the variance of the residuals is approximately constant across different levels of fitted values. The spread of the residuals should remain reasonably similar as predicted charges increase.
 2. **Q-Q Plot + Jarque-Bera test:** Q-Q stands for **Quantile-Quantile** that compares, theoretical quantiles from a normal distribution, observed/sample quantiles of the residuals. The red diagonal line represents what we would approximately expect if the residuals followed a normal distribution.
 - The Q-Q plot provides a visual assessment. The **Jarque-Bera test** provides a formal statistical test of residual normality. There is strong statistical evidence that the regression residuals are not normally distributed.
 3. **Variance Inflation Factor (VIF):** **VIF** is used to check multicollinearity among the predictor variables in a regression model. Multicollinearity means that two or more predictors contain similar/overlapping information.